Mohamed Fathallah

Full-Stack Engineer

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PROFILE

Full-stack engineer with a master's-level degree (Bac+5) and close to three years building production web applications, including 10 months as Team Lead of a six-engineer team. Back end in Java and Spring Boot, front end in React and TypeScript, with hands-on experience in real-time data (Kafka, Redis, WebSocket) and applied machine learning. Involved across the full lifecycle, from specifications and estimation to testing, code quality, documentation and client demos, in Agile.

EXPERIENCE

Full-Stack Engineer, AI Mobility Solutions (AIMS) — Jan 2026 – Present

Sfax, Tunisia

Custom business intelligence platform for an enterprise client in the mobility sector. Leading a team of 6 engineers, in Agile.

- **Front end:** Designed and built the interface with **React 19 / TypeScript**, using **TanStack Router/Query** for routing and data fetching, **Zustand** for state, **Tailwind CSS** for styling and **ApexCharts** for interactive dashboards, with **Zod** for data validation.
- **Back end and security:** Implemented business logic and REST APIs (70 endpoints across 14 controllers, with 46 method-level access rules) with **Java / Spring Boot**, secured with **Spring Security / JWT** for authentication and role-based access control.
- **Data layer:** Modelled and aggregated ~50,000 rows over two years in **PostgreSQL / MongoDB**, with server-side pre-aggregations via **Cube.js** to serve interactive dashboards.
- **Real time and performance:** Connected Java and Python services with **Kafka**, added system-wide caching with **Redis** and pushed live KPIs to the interface over **WebSocket**. Read endpoints answer in 4–15 ms at the median and every page loads in under 0.55 s (production build, local measurements).
- **Automated reporting:** Built a reporting engine with scheduled CSV/PDF exports sent by email, replacing a manual process that took up to a full working day per report.
- **Partner data API (MaaS):** Delivered an API for external partners on top of the Cube data layer: 3 partner endpoints and 12 admin endpoints, token authentication, per-client rate limits (HTTP 429 with Retry-After), revocation effective on the next call, call logging and a **React** admin console. Adds about 12 ms to a Cube query at the median, with 96.6% line coverage on the back end.
- **AI services (in progress):** Designing the Phase 2 AI module for transport-plan planning, alongside a data engineer: a **LightGBM** model (P50/P90 quantiles) for travel-time anomaly detection and forecasting, a deterministic **what-if simulator** that evaluates a modified plan before rollout (vehicle loads, passengers left behind, waiting time, buses needed), and a recommendation engine ranking the best adjustments. Served through **Python / FastAPI**, with **MLflow / MinIO** for model versioning and promotion.
- **Quality and delivery:** Built the CI/CD chain with **GitHub Actions, SonarQube and Trivy**. Back end covered by 745 passing tests (**Spring Boot Test, Karate** API tests, **JaCoCo**: 78% line coverage) and front end by 1,000+ **Vitest / Testing Library** tests (62% line coverage), plus **Playwright** (E2E) and **k6** (load). ESLint and TypeScript at zero errors.
- **Deployment:** Containerised the project with **Docker / Docker Compose** and deployed it behind **Nginx** on an **OVH VPS**.
- **Leadership and client:** Led a team of 6 engineers through code reviews and shared best practices. Turned business needs into technical specifications, estimated modules and ran demos and retrospectives.

Full-Stack Engineer, Tunisian Space Association (UDENE project) — Jan 2024 – Dec 2025

Sfax, Tunisia

Geospatial web platform for a European Earth observation project (UN SDG 11), serving 200+ researchers, in an international Agile team across Türkiye and Serbia.

- **Front end:** Main front-end contributor. Built the application with **React / TypeScript**, using **Redux Toolkit (RTK Query)** for state and API calls, **React Router** for navigation and **Material UI** for the interface.
- **Experiment workflow:** Designed and built the experiment feature end to end, front and back end, in two modes. In manual mode, the user selects an area, a date range (bounded by each dataset's available dates) and a model. In automatic mode, a **GPT** model drives the run from a free-text description of the idea. Developed the run endpoints in **Node.js / Express** and the per-user experiment history, and displayed the similar locations returned by the external datacube / ML service (similarity score and rank) on an **OpenLayers** map.
- **Satellite index viewer:** Built the interactive map with **OpenLayers WebGL**, rendering multi-band **GeoTIFF** imagery in the browser. Spectral indices (NDVI, NDWI, NDBI, NBR, EVI, SAVI) are computed per pixel on the GPU, with custom colour ramps and a live legend.
- **Formula engine:** Wrote a recursive-descent parser in TypeScript that works with **mathjs** to turn index formulas stored in **MongoDB** into WebGL expressions. New indices can be added without code changes.
- **Global explorer:** Integrated 9 **Copernicus / Sentinel Hub WMS** catalogues (32 layers, including Sentinel-2, Landsat 8-9, Sentinel-5P and ERA5), with time-dimension handling and zoom-based rendering.
- **Back end:** Developed **Node.js / Express / TypeScript** services on **MongoDB / Mongoose**: a product-index management API, user administration with role-based access and a feedback module. Extended **JWT / Passport** authentication with an email verification-code flow (**Nodemailer**) and route guards.
- **Admin tools and dashboard:** Built an admin form builder with drag-and-drop (**dnd-kit**) and versioned configurations, plus a **Recharts** dashboard of experiment activity.
- **Collaboration:** Worked daily in Agile with **Git/GitHub**, alongside teams in Türkiye and Serbia.

TECHNICAL SKILLS

Back end: Java, Spring Boot, Spring Security / JWT, REST API design, Clean Architecture, Node.js, Express, Passport, FastAPI

Front end: TypeScript, JavaScript, React 18/19, Next.js, Redux Toolkit (RTK Query), TanStack Router/Query, Zustand, Material UI, Tailwind CSS, ApexCharts, Recharts

Geospatial: OpenLayers (WebGL), GeoTIFF, WMS, Copernicus / Sentinel Hub, proj4, mathjs

Data / ML: LightGBM, scikit-learn, MLflow, MinIO

Testing & quality: Spring Boot Test, Karate, JaCoCo, Vitest, Testing Library, Zod, code reviews, SonarQube, Trivy

Databases & data: PostgreSQL, MongoDB / Mongoose, Redis, Cube.js; modelling, aggregation and data delivery

CI/CD & tools: Git, GitHub, GitHub Actions, Docker, Docker Compose, Nginx, OVH VPS, Cloudflare

Messaging & real time: Kafka, WebSocket

Methodologies: Agile Scrum, estimation, technical specifications, client demos, retrospectives, technical documentation

SOFT SKILLS

Leadership and mentoring: led 6 engineers, ran code reviews and shared good practice within the team

Problem solving: turned complex needs into working solutions, from GPU-based satellite index computation to replacing a day-long manual reporting process

Ownership and autonomy: took modules end to end, from specification through to production support

Client communication: ran workshops and demos, explained technical choices to business teams

Planning and estimation: broke requirements into specifications, estimated modules and followed Agile rituals (planning, demos, retrospectives)

Quality mindset: code reviews, automated testing and quality gates (SonarQube, Trivy) built into the delivery chain

Adaptability and fast learning: moved between two different stacks and domains (Earth observation, mobility BI, applied ML) and became productive quickly

Intercultural and multilingual collaboration: two years working daily with teams in Türkiye and Serbia, working in Arabic, French and English

EDUCATION

Engineer's Degree in Computer Science, International Institute of Technology, Sfax **2021 – 2024**

Preparatory Cycle, Faculté des Sciences de Sfax **2018 – 2020**

LANGUAGES

Arabic (native) · French (C1) · English (B2)